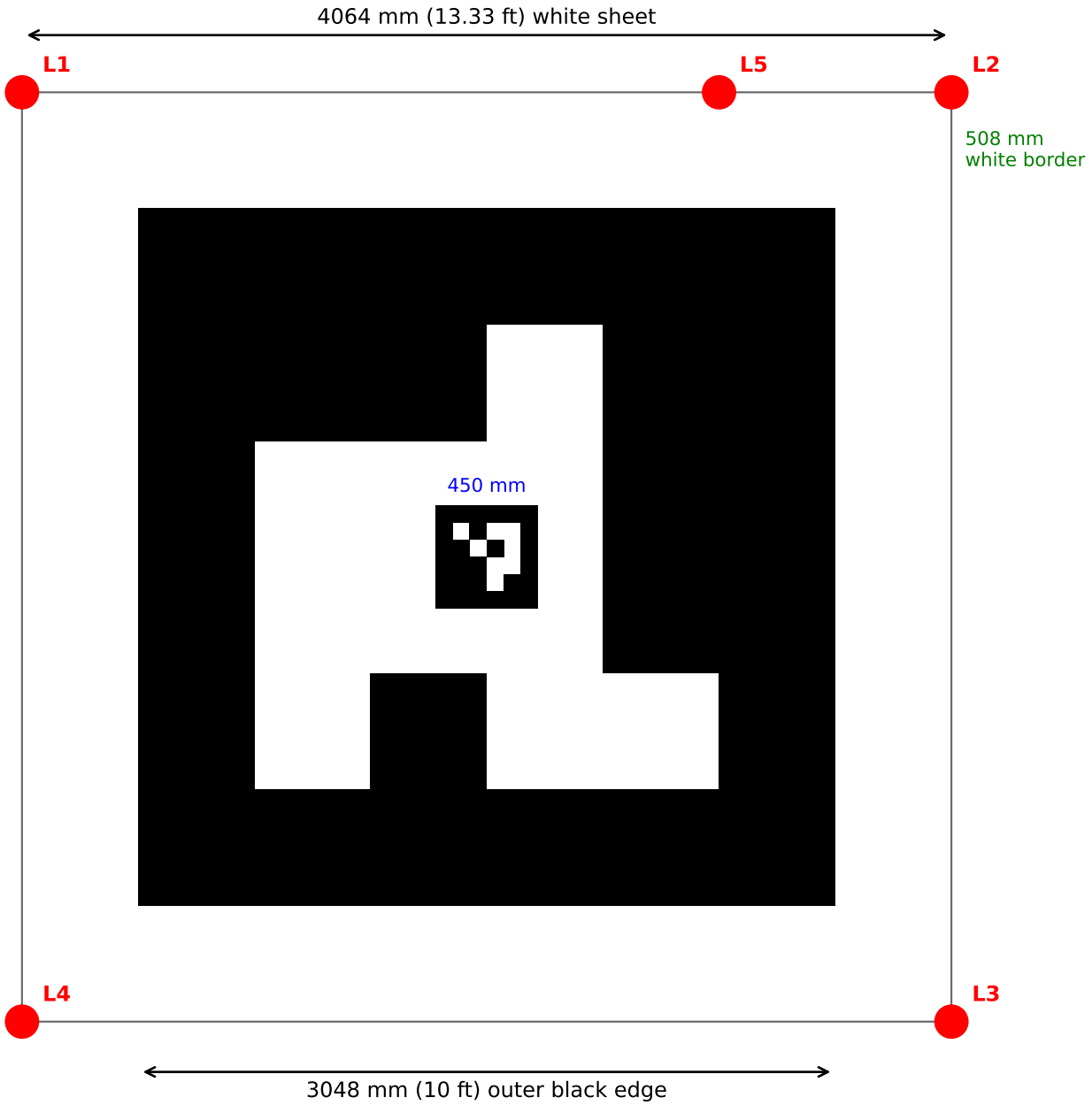
10 ft Nested Landing Pad - Print & Install Guide

Outer ArUco DICT_4X4_50 ID 45 (3.048 m) + inner ID 0 (0.450 m) | TOP edge = forward/north reference

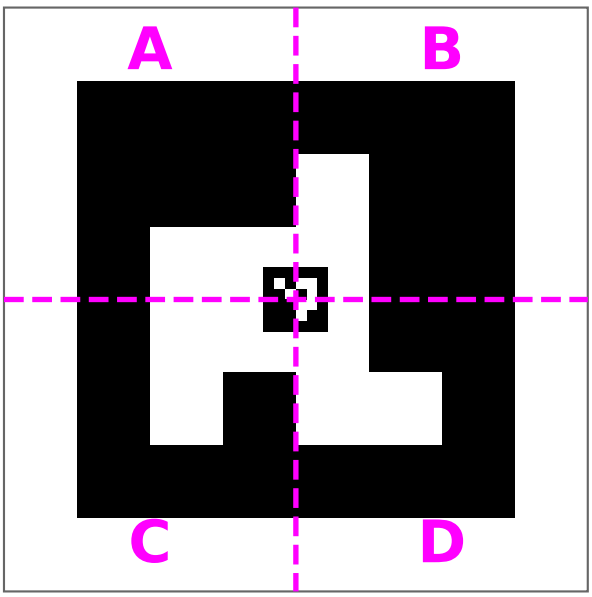


Part	Size
Whole white sheet	4064 x 4064 mm
White border (quiet zone)	508 mm each side
Outer marker ID 45 (black edge)	3048 mm, cell 508 mm
Inner marker ID 0 (black edge, centred)	450 mm, cell 75 mm
Red dots L1-L5	Night IR LED positions (outside the print)

Option B: print in 4 panels (2032 x 2032 mm each)

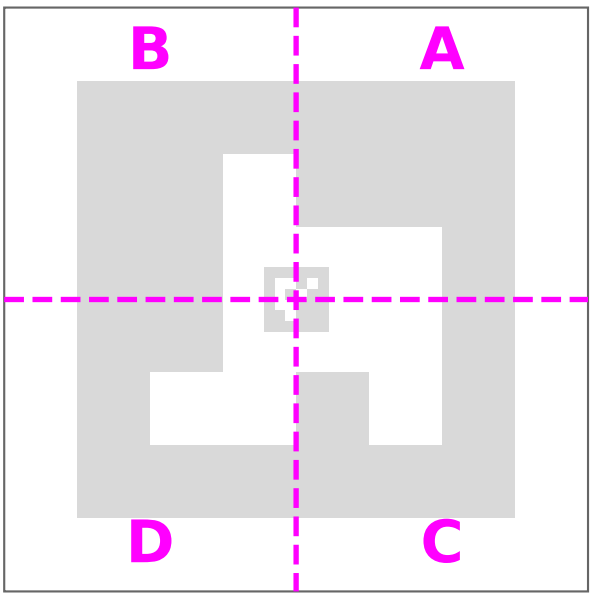
Use when the flex machine is narrower than 4.1 m. Seams fall exactly on cell edges.

TOP edge



FRONT (printed side up)

TOP edge



BACK (printed side down) - tape here

Files: PRINT_panel_A_top-left, B_top-right, C_bottom-left, D_bottom-right (2032 x 2032 mm).

1. Printer: 100 % scale, NO fit-to-page, NO bleed, NO border/crop marks, matte flex.
2. Measure every panel: 2032 mm (+/- 10 mm). Write A/B/C/D and an arrow to TOP on the back.
3. Lay the 4 panels FRONT side up exactly as the left picture: A B on top, C D below.

   Check the black shapes continue across both seams.
4. Flip the whole layout over, one seam at a time, so the BACK is up (right picture:

   now B A on top, D C below). Keep the TOP arrows pointing the same way.
5. Butt the edges tight (no gap, no overlap), tape the full seam length on the back

   with 75-100 mm cloth/duct tape. Tape the centre crossing twice.
6. Turn the pad face up. No white or black gap > 5 mm at the seams.
7. Eyelets every 1 m along the outer edge, within the outer 100 mm of white only.

Printing checklist (flex / banner)

Material

- Matte white flex/vinyl (NOT glossy). Glossy black turns white in sunlight.
- 440-510 GSM frontlit flex is fine. Pure black ink, solid fill (no halftone dots).

File

- Option A: PRINT_nested_pad_FULL_4064x4064mm.pdf (one piece, needs a >= 4.1 m machine).
- Option B: 4 x PRINT_panel_*_2032x2032mm.pdf (join as on page 2).
- Vector PDF, exact size in the file. Scale 100 %. Do not add logos, text, crop marks or border.

After printing - measure before paying

- Whole sheet 4064 mm (+/- 20 mm).
- Outer black square 3048 mm (+/- 15 mm) and exactly centred (508 mm white each side).
- Inner black square 450 mm (+/- 5 mm) in the exact centre.
- Corners square: both diagonals equal within 20 mm.
- If the printed size is different, measure the OUTER black edge and INNER black edge
  and send both numbers: the software needs the real sizes.

Installing the pad on the ground

Site

- Flat, open ground. No poles, trees, people or vehicles casting shadow on the pad.
- At least 10 m clear around for takeoff/landing. Away from metal sheds (compass).

Laying out

1. Clear stones and sticks. The pad must lie flat with no wrinkles.
2. Lay the pad face up. Mark on the back which edge is TOP (row 1).
3. Point the TOP edge toward a fixed reference (e.g. north) and write it down.
4. Pull the sheet tight diagonally, corner to corner.
5. Stake all 4 corners first, then every ~1 m along all edges (ground pegs through eyelets).
6. Add sand bags or weights on the white border if the ground is soft or windy.

   Propeller downwash lifts a loose 4 m sheet - it can wrap the drone.

Checks before flight

- Walk around: no loose edge, no fold, no dirt or water film on black cells.
- Hold a phone 3-4 m up (pole/terrace) and photograph: the whole pad incl. white border visible.
- Keep the white border free of shoes, tools, cables and LED boxes.

Sun and shadow

- A shadow across the pad breaks detection (simulation). Prefer times when the drone's shadow falls beside the pad, not on it.

Night: 5 IR LED beacons (L1-L5)

Positions (see page 1)

- L1 top-left, L2 top-right, L3 bottom-right, L4 bottom-left sheet corners.
- L5 on the TOP edge, 3048 mm from L1 (1016 mm from L2). Makes the pattern asymmetric.
- Place each LED box on the ground just OUTSIDE the corner, never on the sheet.

Hardware (per LED)

- 940 nm (or 850 nm) 3 W IR emitter or 12 V CCTV IR illuminator, wide beam 90-120 deg.
- Constant-current DC driver (700 mA for 3 W). No PWM dimming.
- IP65 box with clear window, LED pointing straight up, levelled with a bubble level.
- One 12 V 20-30 Ah LiFePO4 battery for all 5, fused. Cables along the edges, taped flat.

Checks

- Phone camera shows each LED as a purple/white dot.
- Never look into a lit LED from close range (invisible but harmful).
- Hold the drone (props off) 2-3 m above the pad at night and record images: 5 bright dots and the small inner marker (lit by the drone IR floodlight) must be visible.

Software status

- Today the daemon uses only inner ID 0 on the 640x400 mono camera.
- Outer ID 45, 1280x800 mode and IR beacon detection must be added before long-range or night precision flights.